

No. of Functional Features Included in the Solution

Date	5th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Name	CREDIT CARD APPROVAL PREDICTION
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Collection	Sources and downloads the applicant and credit history datasets	Data Collection (Epic 1)	Done	High
2	Exploratory Data Analysis	Univariate, multivariate, and descriptive analysis of applicant data	Visualizing & Analysing (Epic 2)	Done	Medium
3	Data Cleaning & Merging	Drops duplicates, handles missing values, merges the two datasets	Data Pre-processing (Epic 3)	Done	High
4	Binary Risk Label Engineering	Converts multi-class payment status codes into a safe/risky target label	Data Pre-processing (Epic 3)	Done	High
5	Categorical Encoding & Scaling	Encodes categorical fields and scales numeric fields for modeling	Data Pre-processing (Epic 3)	Done	Medium
6	Multi-Model Training & Comparison	Trains and compares Logistic Regression, Decision Tree, Random Forest, XGBoost	Model Building (Epic 4)	Done	High
7	Web Application Form	Collects applicant details through an HTML form	Application Building (Epic 5)	Done	High
8	Instant Prediction Result	Returns Approved/Rejected with risk category via Flask backend	Application Building (Epic 5)	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Dataset Collection, Data Cleaning & Merging, Binary Risk Label, Multi-Model Training, Web Application Form, Instant Prediction Result)

Additional / Nice-to-Have Features	2 (EDA, Categorical Encoding & Scaling — supporting steps rather than user-facing outcomes)
Features Tested & Verified	8 (all verified — EDA charts confirmed, preprocessing output checked, model comparison table generated, app tested end-to-end with both approved and rejected cases)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Web Application Form
2	Backend / Logic	5	Dataset Collection, EDA, Data Cleaning & Merging, Binary Risk Label Engineering, Categorical Encoding & Scaling, Multi-Model Training
3	Database / Storage	1	CSV dataset files, pickled model/scaler/column files
4	API / Integration	1	IBM Watson Machine Learning (optional, not yet implemented)
5	Security / Authentication	0	Not implemented in current version